

Path Integral Formulation of QM

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Green's function in QM in 1D.

$$G(x', t'; x, t) = \langle x', t' | x, t \rangle \quad (1)$$

This matrix element solves the problem of finding how the wave function evolves with time

$$\psi(x', t') = \int dx G(x', t'; x, t) \psi(x, t) \quad (2)$$

$$(3)$$

Since

$$\psi(x, t) = \langle x | \psi(t) \rangle \quad (4)$$

Given a time evolution operator

$$|\psi(t')\rangle = e^{-\frac{i(t'-t)}{\hbar} \hat{H}} |\psi(t)\rangle \quad (5)$$

We need to allow the particle to make all possible trajectories. Classically our EOM's come from the path that minimises the action. In qm rather than following a path line it 'smears out' into a blur of probabilities around this path with higher probability's closer to the line quantitatively described by a weighting factor.

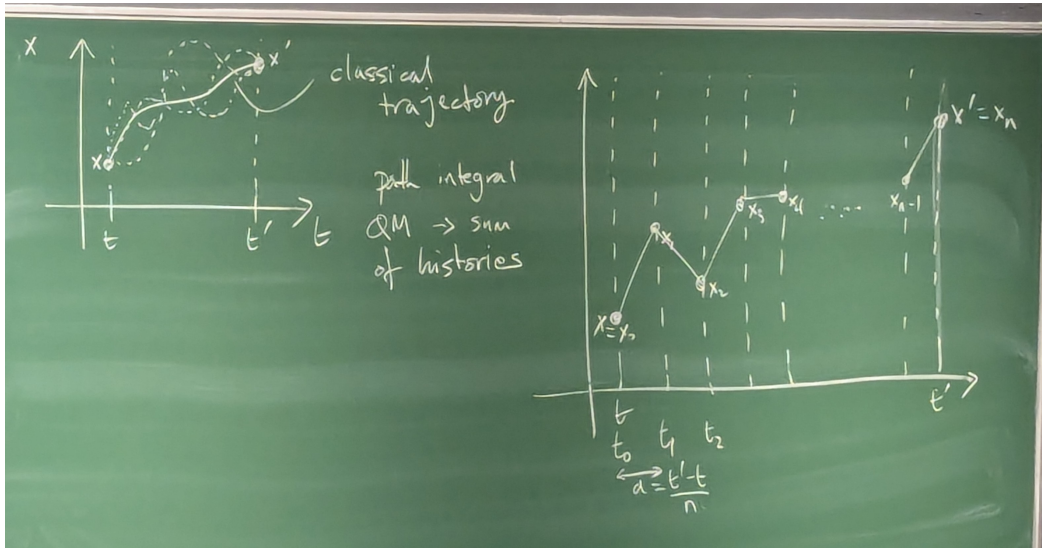


Figure 1: 9.1.jpg

$$G(x', t'; x, t) = \langle x', t' | x, t \rangle = \left\langle x' | e^{-\frac{i(t'-t)\hat{H}}{\hbar}} | x \right\rangle \quad (6)$$

Integrating over each state

$$\int \prod_{k=1}^{n-1} dx_k \left\langle x' | e^{-\frac{ia}{\hbar}} | x_{n-1} \right\rangle \langle x_{n-1} | e^{-\frac{ia}{\hbar}} | x_{n-2} \rangle \langle x_{n-2} | \dots | x_1 \rangle \langle x_1 | e^{-\frac{ia}{\hbar}} | x \rangle \quad (7)$$

To get a more accurate representation $k \rightarrow \infty$.

As we increase the time resolution then $a \rightarrow 0$. Maybe we could taylor expand about a the difference in time.

$$\left\langle x_{k+1} | e^{-\frac{ia\hat{H}}{\hbar}} | x_k \right\rangle \quad (8)$$

with

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}(x, t) \quad (9)$$

$$e^{\frac{-ia\hat{H}}{\hbar}} = e^{\frac{-a\hat{p}^2}{2m\hbar}} e^{\frac{-ia\hat{V}}{\hbar}} - \mathcal{O}(a^2) \quad (10)$$

$$(\mathcal{O}(a^2) \neq 0) \quad (11)$$

Note we need to be careful as we have operators, standard rules for exponentials are invalid if operators don't commute by BCM expansion.

BCM given by

$$e^{h(A+B) + \frac{\hbar^2 X}{2}} = e^{hA} e^{hB} \quad (12)$$

$$1 + h(A+B) + \frac{\hbar^2}{2}(A^2 + AB + BA + B^2), \text{ where we can identify } X = [A, B] \quad (13)$$

NOTE NOT ABELIAN :—